

Source Extraction Checkpoint S2: Amen Bank 2023 Annual Report

This is the second application of the v1 source extraction template, and the first applied to a corporate disclosure document rather than an academic synthetic dataset. The Amen Bank 2023 Annual Report was selected because it is the most authoritative primary source on the institutional context in which our system will operate, and because its genre is structurally complementary to PaySim: real but aggregate where PaySim is synthetic but transactional. This checkpoint records that complementarity explicitly so that Source S3 can be chosen to close the remaining joint gaps.

Source Identification

- **Document:** Amen Bank Annual Report 2023.
- **Publisher:** Amen Bank, S.A., Tunisia (registered office: Avenue Mohamed V, 1002 Tunis).
- **Reporting period:** Fiscal Year 2023 (01 January – 31 December 2023).
- **Approval:** Ordinary General Assembly of 25 April 2024.
- **Auditors:** BDO Tunisie and La Générale d'Audit et Conseil (member of MGI Worldwide). Statutory auditors' report signed 01 April 2024.
- **Format:** PDF (image-based, OCR required).
- **License / access:** Publicly disclosed corporate document, freely redistributable in fair-use academic context.

Document Structure: Tier 1 vs. Tier 2

This source has an internal two-tier trust structure, made explicit by the statutory auditors themselves in their general report:

"Our opinion on the financial statements does not extend to the report of the Management Board and we do not express any form of assurance whatsoever on this report."

— BDO Tunisie & La Générale d'Audit et Conseil, 01 April 2024

This sentence cleaves the 150-page document into two extraction tiers:

- **Tier 1 (audited)** **Balance sheet, income statement, cash flow statement, off-balance sheet commitments, notes to financial statements.** Externally verified by two independent audit firms. High trust on numerical claims.
- **Tier 2 (unaudited management narrative)** **Strategy, activity descriptions, governance commentary, CSR activities, NEXT transformation program, technology deployments, future plans.** No

independent verification. Useful for institutional context but qualitative claims must be cited with hedge language.

Every fact extracted in this checkpoint is tagged [T1] or [T2] in the provenance section.

Trust Profile (7 Dimensions)

Dimension	Grade	Justification
Recency	Strong	FY2023 reported April 2024; ~2-year gap to current date. Structurally current; operational specifics mildly stale.
Authority	Strong / Medium (split)	Strong on T1 (audited financials, two independent firms); Medium on T2 (corporate self-report, no external assurance).
External validation	Medium	Auditor coverage extends only to T1. T2 narrative has no independent validator.
Information Volume*	N/A	Dimension designed for record-count sources (PaySim-style). Does not apply cleanly to a disclosure document. See findings section below.
Geographic relevance	Strong	Tunisia-specific; the bank under study. Strongest possible relevance.
Data type	Medium	Data is real but aggregate (totals, percentages, narrative), not transactional. Calibration-grade, not training-grade.
Reproducibility	Strong	Document is publicly available; quoted facts are verifiable by third parties (access-reproducibility).

Profile summary: 4 Strong / 2 Medium / 1 N/A.

* The "Information Volume" dimension is flagged for revision in template v2 (see the section on v2 Template Findings).

Operation Calibration Pathway

Unlike PaySim, this source does not expose individual transactions and therefore admits no direct (1:1) operation mapping. The contribution takes the form of *indirect calibration mapping* – aggregate figures in the report calibrate parameters of our synthetic generator.

Operation	Calibration source within S2
Retrait	Card-channel withdrawal statistics (Monetics section): ~240k cards in circulation, withdrawal volume +9% year-on-year. Calibrates GAB/card channel only; teller-window retraits remain uncovered.
Versement	Customer deposit composition (Activity section): sight 25.6% / savings 31.6% / term 41.1% / other 1.7%. Calibrates the <i>steady-state</i> composition of customer balances, not the deposit-flow rate.
Virement	Foreign transfers and trade-finance volumes: ~4,195 MTD domiciliation volume; transfers received +23.17%; transfers issued +25.31%. Best-covered of the four operations in this source.
Remise de chèques	Cheque-commission line item exists but no operational breakdown. Persistent gap; same as PaySim.

Coverage Matrix (8 Dimensions)

Dimension	Grade	Note
Amount distribution	Medium	Amounts in TND throughout, but as aggregates only. No transaction-level percentile shape extractable.
Operation type mix	Not covered	Mention of operations is not quantification of their mix. No teller-level breakdown.
Transaction frequency	Low	Annual aggregates with growth rates (+4.1% card withdrawals); cohort-level signal only.
Inter-event time	Not covered	Not exposed in disclosure documents.
Temporal patterns	Medium	Tunisia-specific seasonality: tourism (+57.4% foreign tourists, summer-driven), TRE summer flows, Ramadhan/Aid commercial campaigns, year-end events. PaySim cannot provide this.
Spatial / Branch	High	150 branches across 9 named zones (Tunis I–IV, Cap Bon, Sahel, Sfax, South); 5 Business Centers; 11 Self-Service Areas. Signature contribution.
Channel mix	High	Multi-channel landscape mapped: Amen Mobile, AmenPay, Amennet, teller, GAB/ATM, Self-Service Banking, online cheque remittance, Western Union. Partial quantification (240k cards, 98% contactless, 55 MD online payment volume).
Beneficiary / Emitter	Not covered	No graph-structure data on who-pays-whom. Inverse of PaySim.

Coverage summary: 2 High / 2 Medium / 1 Low / 3 Not covered.

Complementarity with Source S1 (PaySim)

The combined coverage of S1 and S2 inverts cleanly along the matrix: where PaySim is strong, Amen is weak, and vice versa.

Dimension	S1 PaySim	S2 Amen
Amount distribution	Medium	Medium
Operation type mix	High	Not covered
Transaction frequency	Medium	Low
Inter-event time	Medium	Not covered
Temporal patterns	Low	Medium
Spatial / Branch	Not covered	High
Channel mix	Not covered	High
Beneficiary / Emitter	High	Not covered

Joint gaps remaining (motivates Source S3 selection):

- Operation type mix at 4-caisse-op granularity – neither source quantifies this.
- Inter-event time – Amen does not expose it; PaySim signal is weak.
- Beneficiary/emitter patterns specifically for Tunisian banking – PaySim provides graph structure but not Tunisia-shaped.

Anomaly Typology Contribution

PaySim contributed *behavioral* typology (what an anomaly looks like at the event level). Amen Annual Report contributes a structurally different kind of typology – *institutional and regulatory framing* of anomalies. The v1 template currently treats anomaly typology as a single category; this source motivates splitting it (see v2 findings).

Institutional framing provided by S2.

- **Vocabulary:** the bank's term for our object of detection is "*opérations atypiques*" (atypical operations). Adopt this in dashboards and final report.
- **Output pathway:** suspicious operations are reported to **CTAF** (Commission Tunisienne des Analyses Financières) via the **GOAML** module. The Decision Service output schema should anticipate compatibility with this reporting protocol.
- **Regulatory frame:** BCT Circular 2016-03 (Core Indicator approach for operational risk capital) requires banks to reserve capital equal to 15% of the three-year average Net Banking Income against operational risk. Establishes institutional weight of the risk category our system addresses.
- **Pain point to avoid:** the bank's recently deployed AI-based AML system was, by its own description, intended "*to increase the efficiency of its AML/CFT mechanisms by reducing false*

positives." False-positive minimization is therefore a known field problem in this institution; our system should be designed against it from day one.

- **Standards alignment:** FATF requirements and "international best practices." Citing these in the final report supports credibility.

Institutional positioning of the project.

The report describes an existing AI AML system already in production at the bank. This raises the question of where our project sits relative to it. The cleanest positioning, based on a side-by-side read of the annual report and the supervisor's stage subject, is:

"The existing Amen Bank AI AML system performs bank-wide compliance reporting on transaction streams (compliance-officer audience, quarterly tempo, CTAF/regulatory output). The system developed in this internship performs real-time decision support for branch supervisors on teller operations (four caisse operations, dual-actor model including the bank employee, near-real-time latency, LSTM sequence-aware). Same regulatory umbrella; different tools."

This sentence is recommended for use in the rapport's introduction and in the soutenance defense.

Institutional home.

Within the Risk Management Division's five risk categories (credit, market, liquidity, operational, LAB/FT), the project lives precisely at the intersection of **Operational Risk** (fraud, processing errors) and **LAB/FT** (smurfing, atypical operations). The other three categories are out of scope.

| Archetype Calibration

Client archetype priors (from the deposit-by-economic-agent breakdown, Activity section, T1):

- Institutional: 6% (2023), down from 10% (2021)
- Private businesses: 33% (2023), up from 27% (2021)
- Individuals & miscellaneous: 61% (2023), stable

Maps onto the synthetic generator's archetype prior: big-business $\approx$ 6–10%, small-business $\approx$ 25%, individual $\approx$ 61% (further split among salaried, student, retiree, TRE per the strategy section's segmentation; TRE added as a sixth archetype based on its operational distinctiveness).

Employee structural priors (from the HR section, T2 but with audit-corroborated headcount in financial notes):

- 1,139 total employees; 1,133 in active employment
- 922 in the executive / senior management category ($\approx$ 85.3% management ratio)
- 736 holding higher-education degrees

- Network assignment: 685 (60.5%) vs. head office: 448 (39.5%)
- Gender split: 61% male / 39% female
- 57 new hires in 2023 (10 permanent + 47 apprenticeship)

Branch-network priors:

150 branches in 9 geographic zones (Tunis I–IV, Cap Bon, Sahel, Sfax, South); 5 Business Centers (Mohamed V, Berges du Lac, Tunis South, Sousse, Sfax); 11 Self-Service Areas. Calibrates the branch_id distribution and zone-stratification of the simulator.

Provenance (Minimal Citation Plan)

Full per-fact page-numbered citations are not produced for this source. Instead, three pieces of provenance are recorded:

1. **Section tag per fact:** every extracted fact in this checkpoint is tied to a named report section (Activity, Monetics, HR, Risk Management, AML/CFT, Compliance Control, Strategy, Statutory Auditors' Report).
2. **Tier marker** (**T1** / **T2**) attached to each fact, per the document-structure subsection above.
3. **Verbatim preservation** of the small number of load-bearing quotes (auditor disclaimer; AI AML deployment sentence; project-positioning excerpt).

Verbatim-preserved quotes.

- Auditor disclaimer (T1 boundary signal): see Document Structure preamble.
- AI AML deployment (T2 institutional context): *"During 2023, the bank successfully deployed a new AI-based solution to increase the efficiency of its AML/CFT mechanisms by reducing false positives, in-depth analysis of high-risk transactions and behaviors, and improving the quality of KYC data."*

v2 Template Findings Generated by S2

Three structural revisions to the source extraction template are motivated by this source. Logged for application after the next extraction.

1. **Generalize "Sample size" to "Information Volume":** rename the dimension and broaden its semantics to cover both record-count sources (datasets like PaySim) and fact-density sources (corporate and regulatory documents). Avoids splitting into two dimensions; keeps the template at 7.
2. **Split "Operation Mapping" into two modes:** *Direct* (1:1 transactional, dataset-style, PaySim) and *Indirect* (aggregate calibration mapping, disclosure-style, Amen). Both are valid contributions and should be recorded explicitly.
3. **Split "Anomaly Typology" into two sub-categories:** *Behavioral shapes* (what does an anomaly look like at the event level – PaySim's contribution) and *Institutional framing* (regulatory

vocabulary, output pathways, capital requirements, known pain points – Amen's contribution).
Sources may contribute to either or both.

Standing v2 findings (unresolved from prior sources).

The operation-sequence/transition dimension flagged during the PaySim extraction remains pending.